

MONEY CARD PLATFORM — COMPLETE DEVELOPMENT ROADMAP

Fast-Track Schedule: 3 Milestones / Day (Dev 2) + 3 Days Post-Merge QA Window

Document Version: 4.1.0 (Optimized Compact Layout)

Source of Truth: M0 V10 Frozen Contract • [MONEY_CARD_COMPLETE_API_CONTRACT_V10.pdf](#)

Working Standard: 8 Productive Engineering Hours = 1 Working Day • 5 Days / Week

Backend Execution Pace: 3 Milestones per Day for Developer 2 • Backend Delivery in 7 Working Days

Dedicated Post-Merge QA: 3 Working Days for Joint Integration, Concurrency & Release Testing

Total Project Timeline: 10 Working Days (2 Calendar Weeks)

1. Project Summary & Time Required

Platform / Track	Developer	Status	Time Completed	Time Remaining	Total Time
Backend (Node.js/Prisma)	Developer 2	Starting from Scratch	0 Days (0h)	7 Working Days (56h)	7 Working Days (56h)
Web Frontend (React/TS)	Developer 1	91.3% Complete	21 Working Days (168h)	1 Working Day (8h)	22 Working Days (176h)
Staff Mobile App (Flutter)	Developer 1	66.1% Complete	10.25 Working Days (82h)	2 Working Days (16h)	12.25 Working Days (98h)
Post-Merge Testing & QA	Dev 1 + Dev 2	Scheduled Post-Backend	0 Days (0h)	3 Working Days (24h)	3 Working Days (24h)
TOTAL PARALLEL LAUNCH	Dev 1 + Dev 2	Fast-Track Build + Joint QA	31.25 Working Days (250h)	10 Working Days (80h)	10 Working Days (2 Weeks)

2. Developer 2 (Backend) Complete Roadmap Table

All 20 backend milestones required to implement the 65 routes in [MONEY_CARD_COMPLETE_API_CONTRACT_V10.pdf](#) (Paced at 3 Milestones / Day):

Milestone	Task Name	Time Required	Dependencies	Scope / Deliverable
D2-M1	Backend Foundation & Architecture	2.5 Hours (0.33 Day)	None	Setup Express, TypeScript, .env, logger, global error envelope middleware, mount /api/v1 router.
D2-M2	Database Schema & Prisma Setup	2.5 Hours (0.33 Day)	D2-M1	Connect PostgreSQL, 12 Prisma models & relations, run migrations, seed DB from TEST_DATA.json.
D2-M3	Authentication Module	3.0 Hours (0.38 Day)	D2-M2	Login (POST /auth/login), JWT access token (15m), refresh tokens, logout, GET /auth/me, password reset flow.
D2-M4	Authorization & RBAC Middleware	2.5 Hours (0.33 Day)	D2-M3	Role guards (SUPER_ADMIN, ORG_ADMIN, STAFF), 20 permissions evaluator, tenant org & branch isolation.
D2-M5	Organizations Management APIs	2.5 Hours (0.33 Day)	D2-M4	Super Admin tenant listing, provisioning, detail, name/status updates with inactive cascade.
D2-M6	Plans, Subscriptions & Overrides	3.0 Hours (0.38 Day)	D2-M5	Global plans (NO transaction limits), org assignment, custom quota overrides, direct payments, plan requests.
D2-M7	Branches & Staff Management	2.5 Hours (0.33 Day)	D2-M4	Branch CRUD, staff CRUD, branch assignments replacement, permission replacement, branch limit checks.

D2-M8	Cards Management & Resolution	2.5 Hours (0.33 Day)	D2-M7	Card list with status filters, physical card creation in AVAILABLE state with qrToken, QR resolve, block/unblock.
D2-M9	Card Sessions Lifecycle	3.0 Hours (0.38 Day)	D2-M8	Session creation on AVAILABLE card (card -> ACTIVE), active session list, session detail with balance.
D2-M10	Products Catalog Management	2.5 Hours (0.33 Day)	D2-M7	Branch products list, product creation with multi-select category array (NO tags), price validation, auto InventoryItem init.
D2-M11	POS & Purchase Financials Engine	3.0 Hours (0.38 Day)	D2-M9, D2-M10	Purchase execution in atomic \$transaction: server price lookup, balance & stock check, inventory deduction, Transaction record.
D2-M12	Recharge & Payment Records	2.5 Hours (0.33 Day)	D2-M9	Card session top-up with CASH and manual UPI verification, balance addition, idempotency reference support.
D2-M13	Refund & Session Settlement	2.5 Hours (0.33 Day)	D2-M11, D2-M12	Session return: calculate balance, record REFUND transaction, zero balance, session -> SETTLED, card -> AVAILABLE.
D2-M14	Inventory & 3-Column CSV Import	3.0 Hours (0.38 Day)	D2-M10	Stock list, stock adjustments (prevent negative stock), 3-col CSV template (itemName,category,price), 2-phase preview & commit.
D2-M15	Analytics & Performance Metrics	2.5 Hours (0.33 Day)	D2-M11, D2-M12	Real-time aggregated metrics (recharge/purchase/refund volume, branch comparison, peak hours), PDF export streaming.
D2-M16	Formal PDF Reports Engine	2.5 Hours (0.33 Day)	D2-M15	Formal reports list and binary PDF streaming download (PDF only per M0 V10).
D2-M17	Public Customer User Portal APIs	3.0 Hours (0.38 Day)	D2-M9	Public card QR resolution (masked data & portal token), portal session access, active balance, transactions, receipts.
D2-M18	Audit Logging System	2.5 Hours (0.33 Day)	D2-M4	Automated audit interceptors for auth events, card state changes, financials, staff assignments, plan changes.
D2-M19	API Contract Parity Verification	4.0 Hours (0.50 Day)	D2-M1 to D2-M18	Systematic verification of all 65 routes against MONEY_CARD_COMPLETE_API_CONTRACT_V10.pdf.
D2-M20	Backend Integration & Postman QA	4.0 Hours (0.50 Day)	D2-M19	Execution of all 65 requests in POSTMAN_COLLECTION.json (100% pass) and Handover to Dev 1.

Developer 2 Total Time: 7 Working Days (56 Hours / 1.4 Working Weeks)

3. Developer 1 (Web Frontend) Roadmap Table

(React • TypeScript • Vite • Mock API & Real API Adapter)

Milestone	Task Name	Time Required	Current Status	Scope / Deliverable
M1	Foundation & Design System	2.0 Days (16h)	■ COMPLETED	Design tokens, dark/light themes, responsive navigation, top header, breadcrumbs, standard UI kit.
M2	Authentication & RBAC Suite	2.0 Days (16h)	■ COMPLETED	Login screens, forgot password modal (neutral response), reset password, change password, profile dropdown, role guards.
M3	Cards Management & Issue	2.0 Days (16h)	■ COMPLETED	Card list, QR barcode viewer modal, manual card creation, CSV template download & import, status filter with Available count.
M4	Sessions Management	1.75 Days (14h)	■ COMPLETED	Active session dashboard, session issuance, branch filtering, live session balance indicators, session details modal.
M5	POS Terminal & Products	2.25 Days (18h)	■ COMPLETED	POS checkout cart, product catalog grid, search/filter, multi-select category tags (string[]), price calculation (no tags).
M6	Payments, Recharge & Refunds	1.5 Days (12h)	■ COMPLETED	Cash top-up modal, manual UPI verification modal, digital receipt modal, session return/refund modal with settlement confirmation.
M7	Inventory & 3-Column CSV	2.0 Days (16h)	■ COMPLETED	Live inventory table, stock adjustments, low stock badges, 3-column CSV template download, branch 2-phase preview & commit.
M8	Analytics & Peak Demand	2.0 Days (16h)	■ COMPLETED	Real-time analytics dashboard, branch performance comparison table, peak hours traffic charts, hourly breakdown, PDF export.
M9	Offline Resilience & Mock Store	1.5 Days (12h)	■ COMPLETED	In-memory mock database store, realistic latency simulation, localStorage state persistence, reset store utility.
M10	Admin Controls & Subscriptions	2.25 Days (18h)	■ COMPLETED	Super Admin tenant org manager, plan comparison matrix, plan upgrade/downgrade request form, limit overrides manager.
M11	Web Polish & Contract QA	1.75 Days (14h)	■ COMPLETED	Automated contract test runner integration, 58/58 test pass verification, zero TypeScript errors, production build optimization.
M12	Web Real Backend Integration	1.0 Day (8h)	■ REMAINING (POST-MERGE)	Switch .env VITE_USE MOCK_API=false to point to live Node.js server (http://localhost:4000/api/v1), smoke test all 65 routes.

Web Frontend Status: 21 Working Days (168 Hours) COMPLETED • 1 Working Day (8 Hours) REMAINING

4. Developer 1 (Flutter Mobile App) Roadmap Table

(Flutter Money card • Android First • Light Mode Only • Minimal UI)

Milestone	Task Name	Time Required	Current Status	Scope / Deliverable
M13	Flutter Foundation & Architecture	1.75 Days (14h)	■ COMPLETED	Light theme, Dio network client with Auth/Error/Mock interceptors, SecureStorage token manager, AppRouter, AppShell.
M14	Staff Authentication & Permissions	1.5 Days (12h)	■ COMPLETED	Login screen, token refresh handling, branch switcher, PermissionGuard, role/permission providers, auth service.
M15	Staff Cards & QR Operations	2.25 Days (18h)	■ COMPLETED	Camera QR scanner (QRScannerView), manual card entry fallback, QR resolve (POST /cards/resolve), card list, Home active session counter.
M16	Payments, POS Checkout & Return	2.75 Days (22h)	■ COMPLETED	Cash recharge screen, manual UPI verification screen (store's physical QR used), POS catalog cart (POSCartProvider), return card & refund screen.
M17	Inventory & Shift Analytics	2.0 Days (16h)	■ COMPLETED	Inventory screen with live stock, quick stock adjustment, AddProductScreen with multi-category selector, staff AnalyticsScreen with volume cards.
M18	Real Backend Integration & QA	1.0 Day (8h)	■ REMAINING (POST-MERGE)	Switch AppConfig useMockApi = false to point to live Node.js server, verify Dio JSON serialization against live API, test error dialogs (ERROR_CONTRACTS.json).
M19	Hardware & Scanner Hardening	0.5 Day (4h)	■ REMAINING (POST-MERGE)	Physical Android device camera optimization for low-light conditions, scan vibration feedback, thermal receipt printer readiness.
M20	Android Production QA & Release	0.5 Day (4h)	■ REMAINING (POST-MERGE)	Release keystore signing, ProGuard rules, Android APK/AAB build generation, offline network error handling validation.

Flutter Mobile Status: 10.25 Working Days (82 Hours) COMPLETED • 2 Working Days (16 Hours) REMAINING

5. Master Parallel Daily Schedule Table

Timeline	Developer 2 (Backend Track)	Developer 1 (Web & Flutter Track)	Daily Target Output
Day 1	D2-M1 (Foundation), D2-M2 (Prisma DB), D2-M3 (Auth)	Flutter Camera QR Scanner Optimization on Android	*Backend running with DB connection & Auth; Flutter camera scanner tested.*
Day 2	D2-M4 (RBAC), D2-M5 (Orgs), D2-M6 (Plans & Overrides)	Flutter Shift Sales Analytics UI Polish & SnackBar Refinements	*Admin, Plans & Orgs ready on Backend; Flutter UI hardened.*
Day 3	D2-M7 (Staff), D2-M8 (Cards), D2-M9 (Card Sessions)	Flutter Offline Test Suite Execution & Token Hardening	*Cards & Sessions live on Backend; Flutter mock suite 100% verified.*
Day 4	D2-M10 (Products), D2-M11 (POS Engine), D2-M12 (Recharge)	Automated Integration Test Suite Preparation for Web & Mobile	*Core Financial & POS engines complete on Backend.*
Day 5	D2-M13 (Refunds), D2-M14 (CSV Import), D2-M15 (Analytics & PDF)	Standby for Backend Financial Endpoints Verification	*CSV Import, Analytics & PDF Reports functional on Backend.*
Day 6	D2-M16 (Reports), D2-M17 (Portal), D2-M18 (Audit Logging)	Web Frontend & Flutter Live API Configuration Scripts Ready	*Public Portal & Reports live on Backend; all 65 routes implemented.*
Day 7	D2-M19 (Parity), D2-M20 (Postman 65 Requests QA)	Pre-merge verification of Web (M1–M11) and Flutter (M13–M17)	*■ BACKEND 100% DELIVERED FOR MERGING!*
Day 8	Joint Backend Support & CORS / Query Logs Verification	Web M12: Real API Integration & 65-Route Contract QA	*Web Frontend 100% Integrated with live backend!*
Day 9	Joint Backend Support & Live PostgreSQL Transaction Monitoring	Flutter M18: Live Integration & Hardware QR Scanner QA	*Flutter Mobile 100% Integrated with live backend!*
Day 10	Final Backend Support, Multi-Tenant Boundary & Concurrency Tests	Flutter M19 & M20: Production Build & Final Release Sign-Off	*■ FINAL SIGN-OFF: Web, Mobile, and Backend 100% Integrated!*

■ Final Summary:

- Total Project Calendar Duration (Parallel): 10 Working Days (2 Calendar Weeks)
- Backend Delivery (Dev 2): 7 Working Days (at 3 Milestones / Day)
- Post-Merge Integration & Testing (Dev 1 + Dev 2): 3 Working Days (Days 8, 9, 10)